

FOOD DELIVERY DATA ANALYTICS

SQL Analysis Report — Data • Visuals • Insights

Dataset: food_delivery_db.food_orders | **Rows:** 100,000 | **Overall revenue:** ■133,597,720 | **Cancelled:** 15,036 (15.04%)

Prepared from the SQL outputs supplied. Where output columns or rows were not supplied, the report notes the limitation.

1. Top-Spending Customers

Objective: Customer spending among delivered orders.

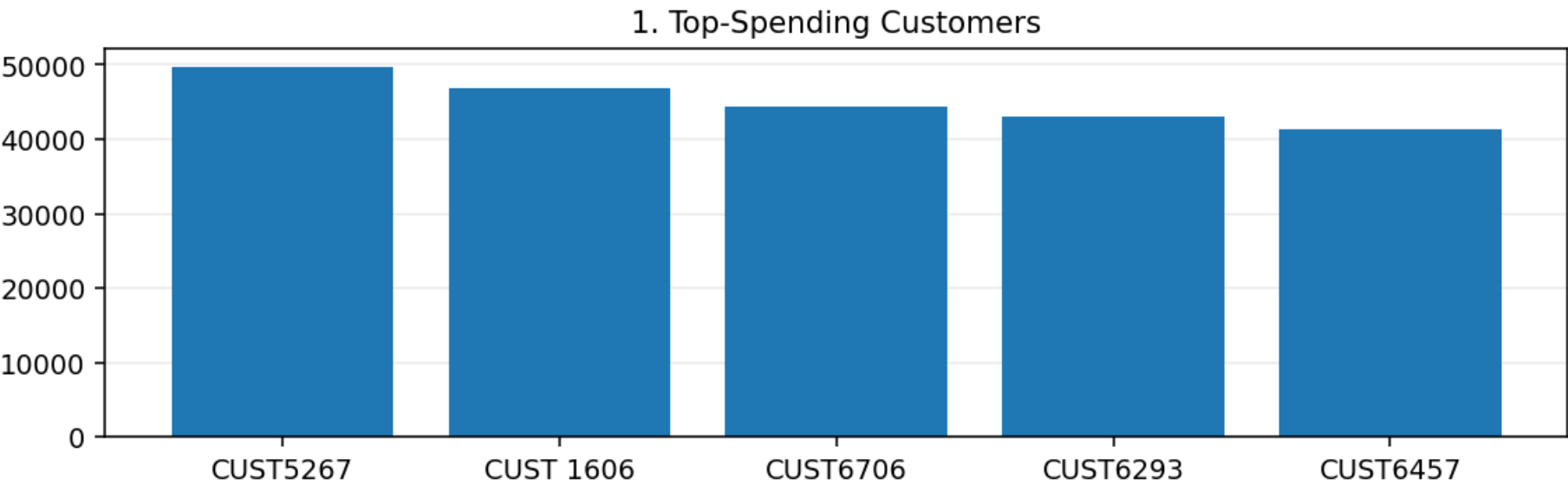
SQL Query

```
SELECT Customer_ID, COUNT(Order_ID) AS Total_Orders, SUM(Final_Amount) AS Total_Spending FROM food_orders WHERE Order_Status='Delivered' GROUP BY Customer_ID ORDER BY Total_Spending DESC LIMIT 10;
```

Data — actual supplied output

Customer ID	Orders	Spending (■)
CUST5267	20	49,683
CUST 1606	20	46,943
CUST6706	18	44,298
CUST6293	21	43,114
CUST6457	20	41,360

Visual



Insights

- CUST5267 leads the supplied rows with ■49,683 spending.
- CUST6293 has the highest order count among the five shown (21).
- Only five result rows were supplied, although the query uses LIMIT 10.

2. Monthly Revenue Trend

Objective: Monthly order count and revenue for records with a usable date.

SQL Query

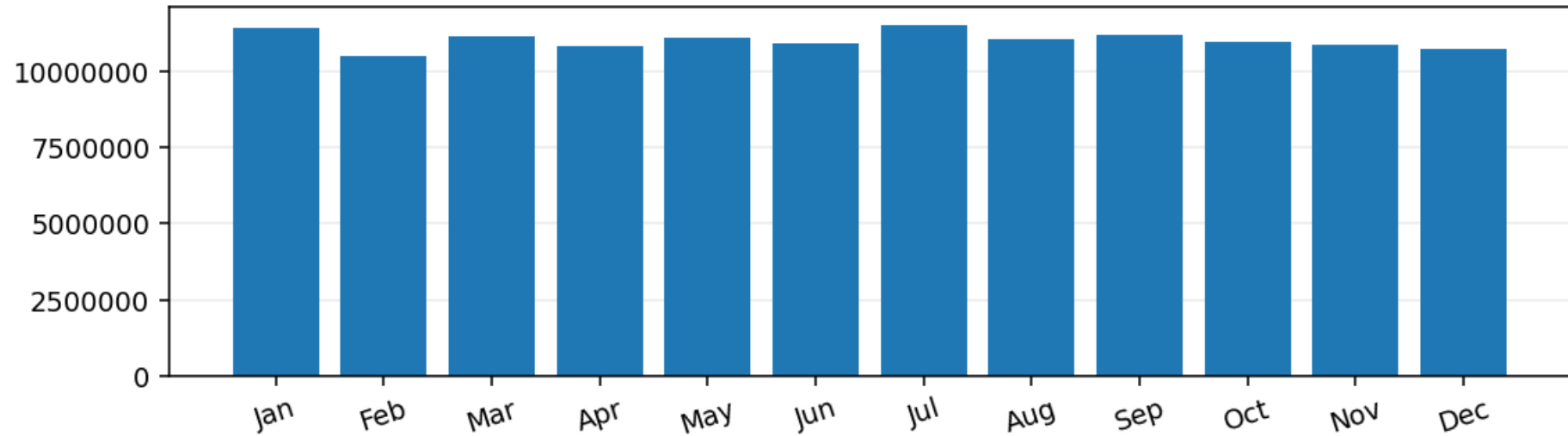
```
SELECT DATE_FORMAT(Order_Date, '%Y-%m') AS Month, COUNT(Order_ID) AS Orders, SUM(Final_Amount) AS Revenue FROM food_orders WHERE Order_Date IS NOT NULL GROUP BY Month ORDER BY Month;
```

Data — actual supplied output

Month	Orders	Revenue (■)
2024-01	8,508	11,410,390
2024-02	7,921	10,496,529
2024-03	8,417	11,130,711
2024-04	8,115	10,848,234
2024-05	8,350	11,089,590
2024-06	8,180	10,922,009
2024-07	8,513	11,537,624
2024-08	8,261	11,060,669
2024-09	8,183	11,192,129
2024-10	8,306	10,989,311
2024-11	8,182	10,860,263
2024-12	8,050	10,715,408

Visual

2. Monthly Revenue Trend



Insights

- July has the highest revenue: █11,537,624.
- February has the lowest: █10,496,529.
- This output covers 98,986 orders; 1,014 rows with missing Order_Date are excluded.

3. Weekday vs Weekend

Objective: Compare order volume and revenue by day type.

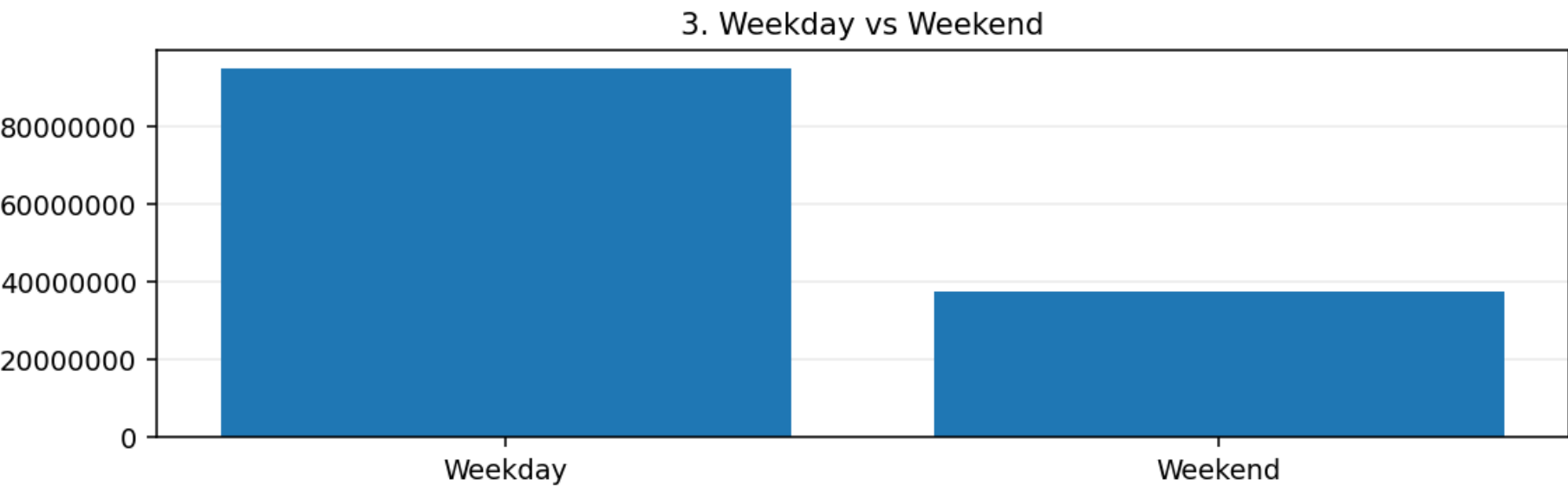
SQL Query

```
SELECT Order_Day_Type, COUNT(Order_ID) AS Total_Orders, SUM(Final_Amount) AS Total_Revenue FROM food_orders WHERE Order_Date IS NOT NULL GROUP BY Order_Day_Type;
```

Data — actual supplied output

Day Type	Orders	Revenue (■)
Weekday	70,618	94,647,453
Weekend	28,368	37,605,414

Visual



Insights

- Weekdays contribute about 71.6% of revenue in this result.
- Weekend revenue contributes about 28.4%.
- Totals cover 98,986 orders with usable dates.

4. Revenue by Cuisine

Objective: Compare order counts and revenue by cuisine.

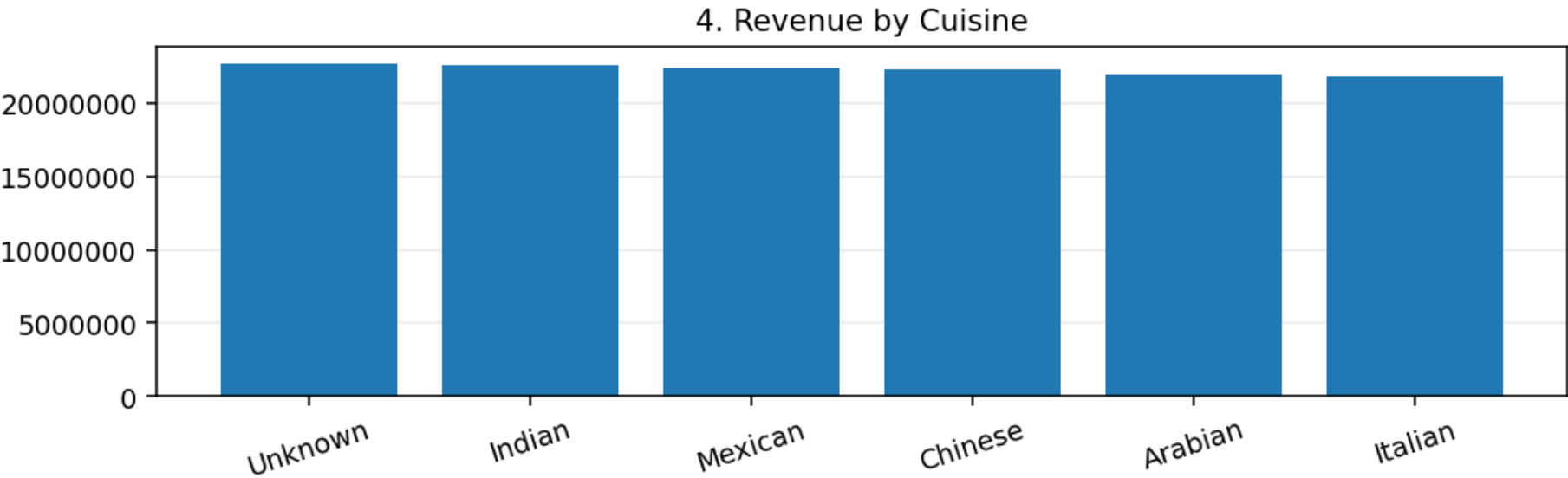
SQL Query

```
SELECT COALESCE(NULLIF(TRIM(Cuisine_Type),''),'Unknown / Missing') AS Cuisine, COUNT(Order_ID) AS Total_Orders, SUM(Final_Amount) AS Total_Revenue FROM food_orders GROUP BY Cuisine ORDER BY Total_Revenue DESC;
```

Data — actual supplied output

Cuisine	Orders	Revenue (■)
Unknown / Missing	16,885	22,696,702
Indian	16,685	22,569,676
Mexican	16,602	22,350,621
Chinese	16,651	22,309,628
Arabian	16,658	21,892,524
Italian	16,519	21,778,569

Visual



Insights

- Unknown / Missing has the highest recorded revenue, ■22,696,702.
- Indian cuisine is next at ■22,569,676.
- Cuisine is missing/blank in 16,885 rows, so category comparisons need caution.

5. Cancellation Rate by Cuisine

Objective: Compare cancellation rate by cuisine.

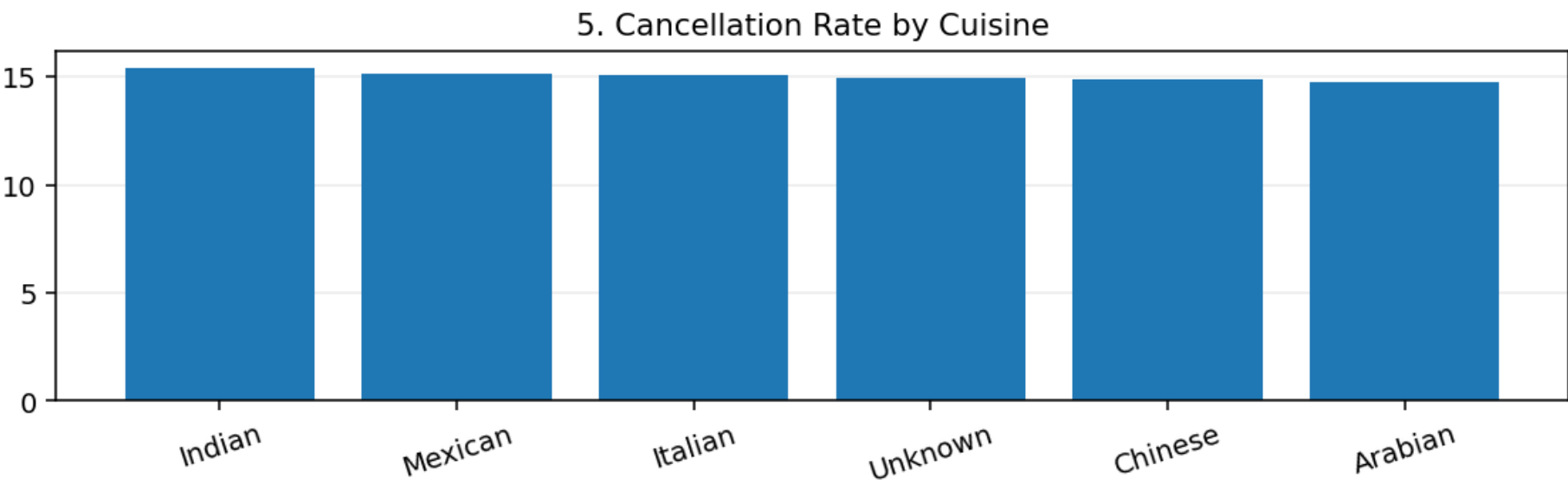
SQL Query

```
SELECT COALESCE(NULLIF(TRIM(Cuisine_Type),''),'Unknown / Missing') AS Cuisine, COUNT(*) AS Total_Orders, SUM(Order_Status='Cancelled') AS Cancelled_Orders, ROUND(100*SUM(Order_Status='Cancelled')/COUNT(*),2) AS Cancellation_Rate_Percent FROM food_orders GROUP BY Cuisine ORDER BY Cancellation_Rate_Percent DESC;
```

Data — actual supplied output

Cuisine	Orders	Cancelled	Rate
Indian	16,685	2,572	15.42%
Mexican	16,602	2,512	15.13%
Italian	16,519	2,491	15.08%
Unknown / Missing	16,885	2,525	14.95%
Chinese	16,651	2,478	14.88%
Arabian	16,658	2,458	14.76%

Visual



Insights

- Indian has the highest cancellation rate: 15.42%.
- Arabian has the lowest: 14.76%.
- The difference between highest and lowest is 0.66 percentage points.

6. Cancellation Reasons

Objective: Break down the reasons recorded for cancelled orders.

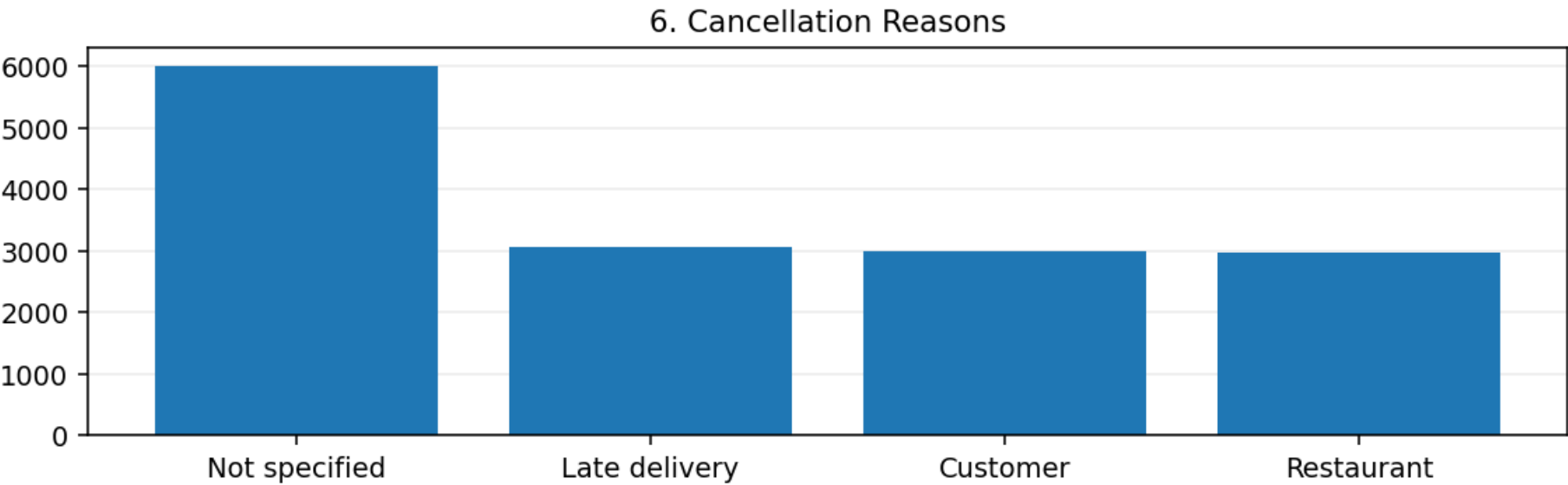
SQL Query

```
SELECT COALESCE(NULLIF(TRIM(Cancellation_Reason),''),'Not Specified') AS Reason, COUNT(*) AS Cancelled_Orders FROM food_orders WHERE Order_Status='Cancelled' GROUP BY Reason ORDER BY Cancelled_Orders DESC;
```

Data — actual supplied output

Reason	Cancelled Orders	Share
Not Specified	6,005	39.94%
Late Delivery	3,059	20.34%
Customer Cancelled	2,993	19.91%
Restaurant Issue	2,979	19.81%

Visual



Insights

- 39.94% of cancellations have no specific reason recorded.

- Late delivery is the largest specified reason (3,059).
- Capture cancellation reasons consistently to support root-cause analysis.

7. City Performance

Objective: Compare order volume, revenue, and average order value by city.

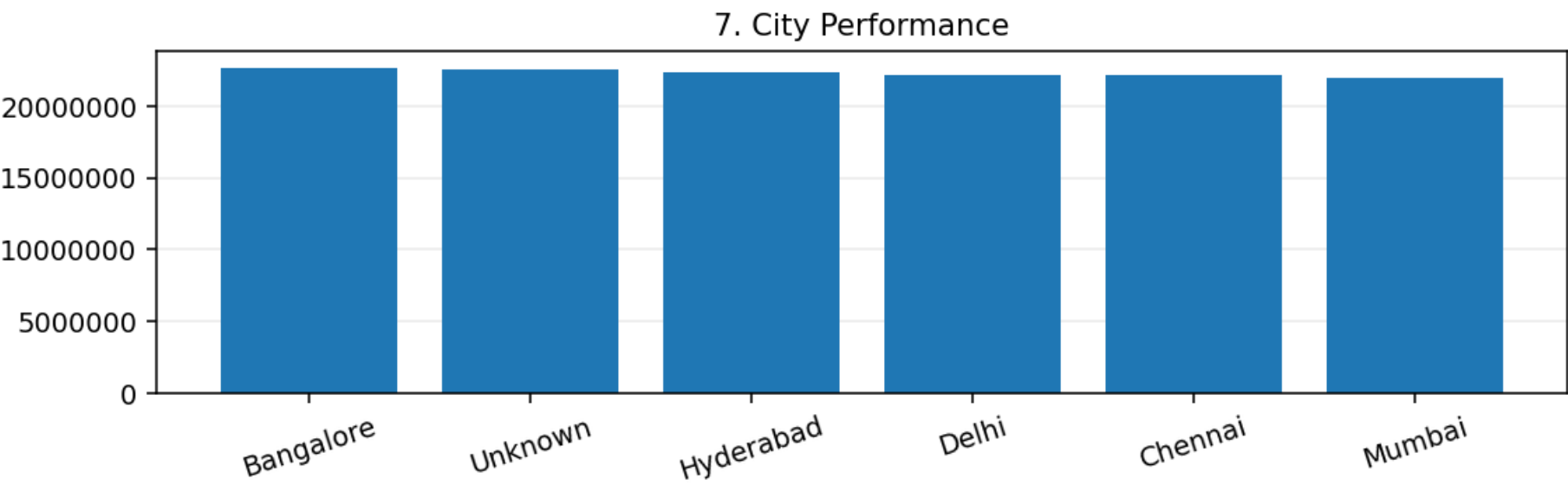
SQL Query

```
SELECT COALESCE(NULLIF(TRIM(City),''),'Unknown / Missing') AS City, COUNT(Order_ID) AS Total_Orders, SUM(Final_Amount) AS Total_Revenue, ROUND(AVG(Final_Amount),2) AS Average_Order_Value FROM food_orders GROUP BY City ORDER BY Total_Revenue DESC;
```

Data — actual supplied output

City	Orders	Revenue (₹)	Avg Order (₹)
Bangalore	16,732	22,639,510	2,013.00
Unknown / Missing	16,726	22,463,657	2,004.43
Hyderabad	16,884	22,331,477	1,990.68
Delhi	16,695	22,100,568	1,996.26
Chennai	16,470	22,098,074	2,016.43
Mumbai	16,493	21,964,434	2,001.68

Visual



Insights

- Bangalore has the highest city revenue: ■22,639,510.
- Chennai has the highest average order value: ■2,016.43.
- 16,726 orders have unknown/missing city information.

8. Payment Mode Analysis

Objective: Compare orders, revenue, and average order value by payment mode.

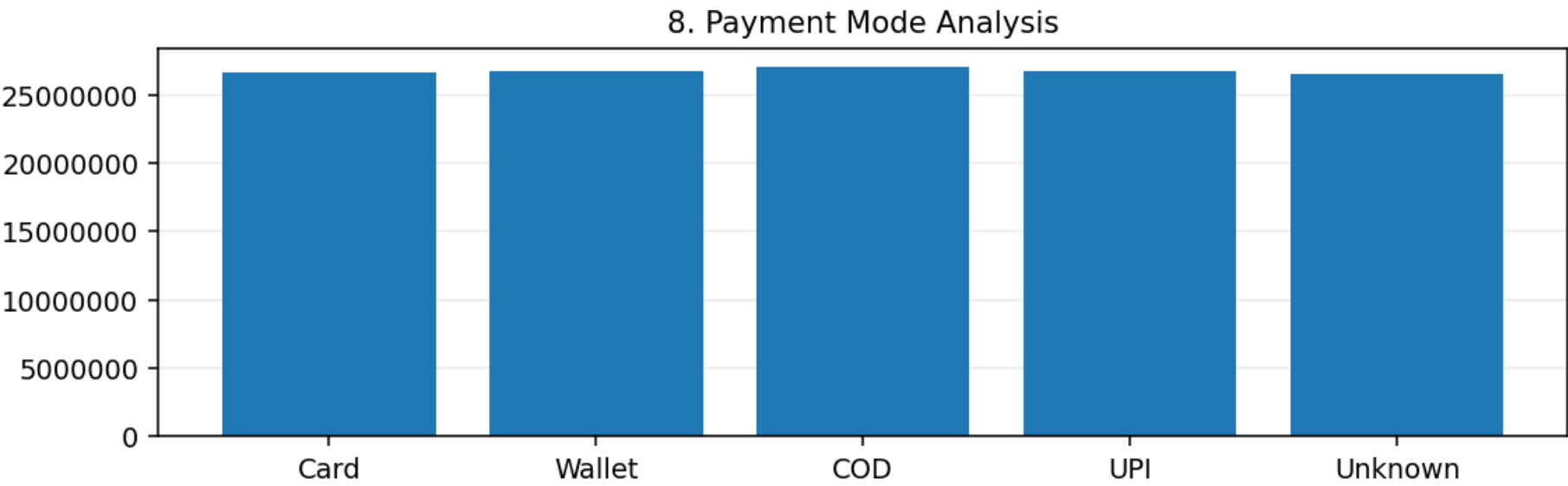
SQL Query

```
SELECT COALESCE(NULLIF(TRIM(Payment_Mode),''),'Unknown / Missing') AS Payment_Mode, COUNT(Order_ID) AS Total_Orders, SUM(Final_Amount) AS Total_Revenue, ROUND(AVG(Final_Amount),2) AS Average_Order_Value FROM food_orders GROUP BY Payment_Mode ORDER BY Total_Revenue DESC;
```

Data — actual supplied output

Payment Mode	Orders	Revenue (■)	Avg Order (■)
Card	20,094	26,590,053	1,990.57
Wallet	20,086	26,732,864	1,993.35
COD	19,977	27,015,584	2,016.84
UPI	19,932	26,701,887	2,015.85
Unknown / Missing	19,911	26,557,332	2,002.36

Visual



Insights

- COD has the highest revenue: █27,015,584.
- COD also has the highest average order value: █2,016.84.
- Payment-mode revenues are close; 19,911 records have unknown/missing mode.

9. Customer Age Group

Objective: Show order distribution across the supplied age groups.

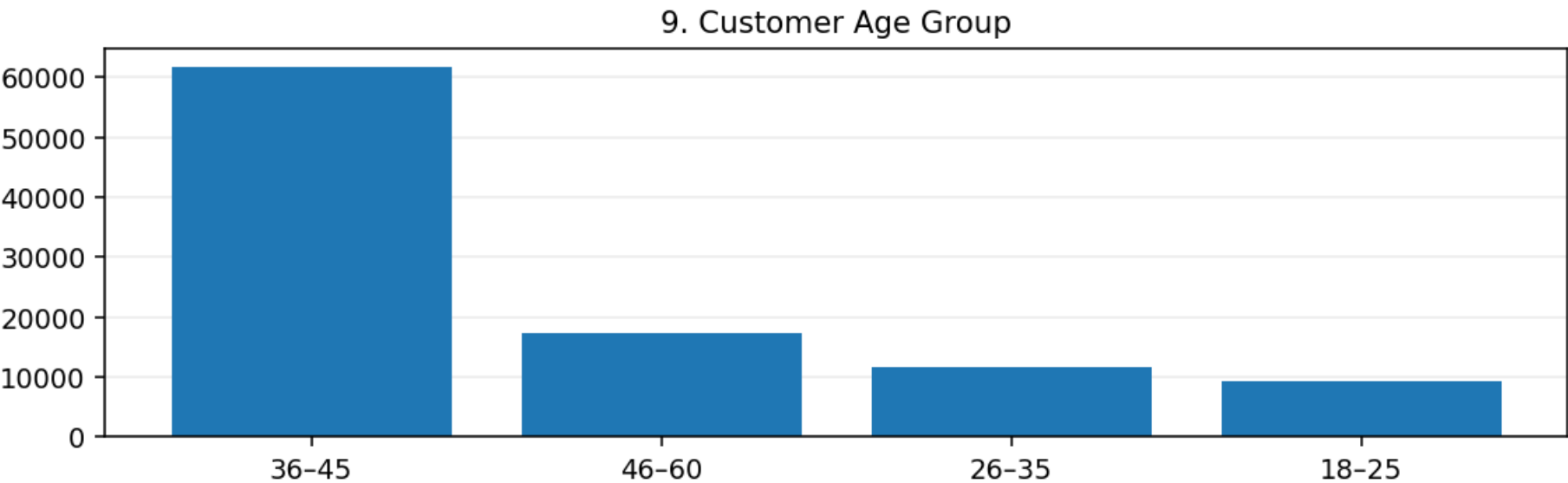
SQL Query

```
SELECT Age_Group, COUNT(Order_ID) AS Total_Orders FROM food_orders GROUP BY Age_Group ORDER BY Total_Orders DESC;
```

Data — actual supplied output

Age Group	Orders
36–45	61,715
46–60	17,386
26–35	11,653
18–25	9,246

Visual



Insights

- The 36–45 group accounts for 61.7% of all 100,000 orders.
- The 46–60 group contributes 17,386 orders.

- Only order counts were provided; revenue and average order value are not reported here.

10. Orders by Customer Gender

Objective: Compare order counts, revenue, and average order value by gender group.

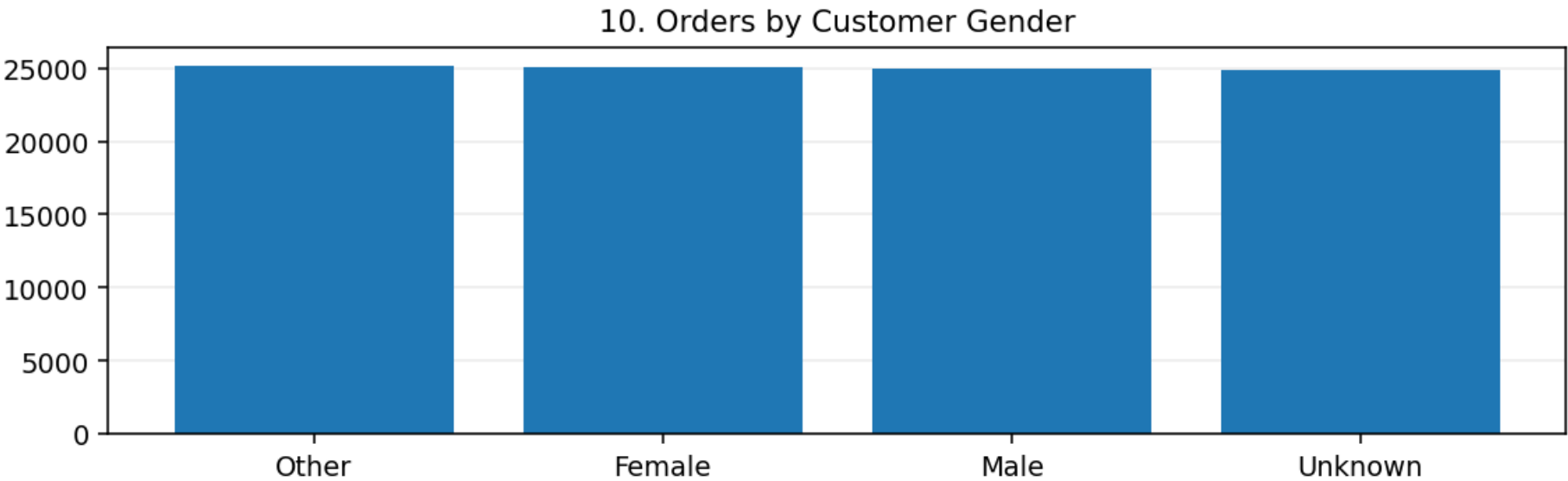
SQL Query

```
SELECT COALESCE(NULLIF(TRIM(Customer_Gender),''),'Unknown / Missing') AS Gender, COUNT(Order_ID) AS Total_Orders, SUM(Final_Amount) AS Total_Revenue, ROUND(AVG(Final_Amount),2) AS Average_Order_Value FROM food_orders GROUP BY Gender ORDER BY Total_Orders DESC;
```

Data — actual supplied output

Gender	Orders	Revenue (■)	Avg Order (■)
Other	25,135	33,530,828	2,012.17
Female	25,031	33,902,176	2,013.55
Male	24,978	33,247,618	1,997.33
Unknown / Missing	24,856	32,917,098	1,991.84

Visual



Insights

- Order counts are relatively evenly distributed across categories.

- Female has the highest revenue and average order value in this output.
- Unknown/missing gender represents 24,856 orders.

11. Delivery Time by City

Objective: Compare delivery duration for delivered orders only.

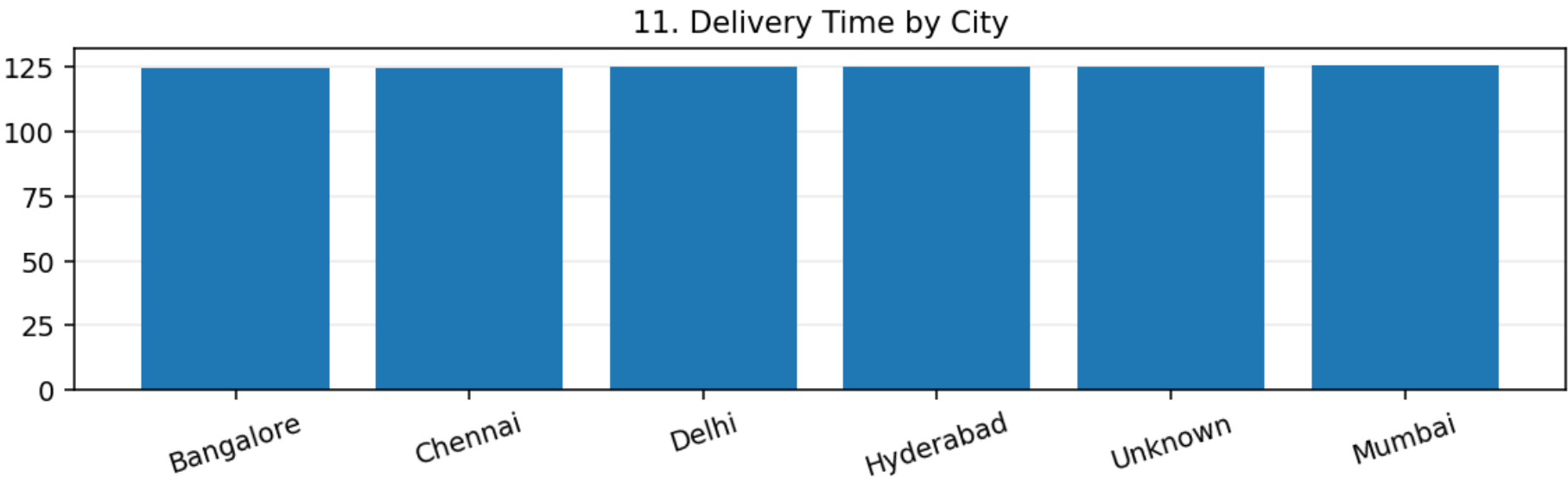
SQL Query

```
SELECT COALESCE(NULLIF(TRIM(City),''),'Unknown / Missing') AS City, COUNT(Order_ID) AS Delivered_Orders, ROUND(AVG(Delivery_Time_Min),2) AS Avg_Delivery_Min, MIN(Delivery_Time_Min) AS Min_Min, MAX(Delivery_Time_Min) AS Max_Min FROM food_orders WHERE Order_Status='Delivered' GROUP BY City ORDER BY Avg_Delivery_Min;
```

Data — actual supplied output

City	Delivered Orders	Avg min	Min	Max
Bangalore	14,231	124.33	20	300
Chennai	13,988	124.34	20	300
Delhi	14,181	125.03	20	300
Hyderabad	14,326	125.18	20	300
Unknown / Missing	14,206	125.19	20	300
Mumbai	14,032	125.79	20	300

Visual



Insights

- Bangalore has the lowest average delivery time (124.33 minutes).
- Mumbai has the highest average (125.79 minutes).
- All cities have a 20–300 minute observed range; review long-duration cases.

12. Restaurant Rating by Cuisine

Objective: Compare average restaurant rating by cuisine.

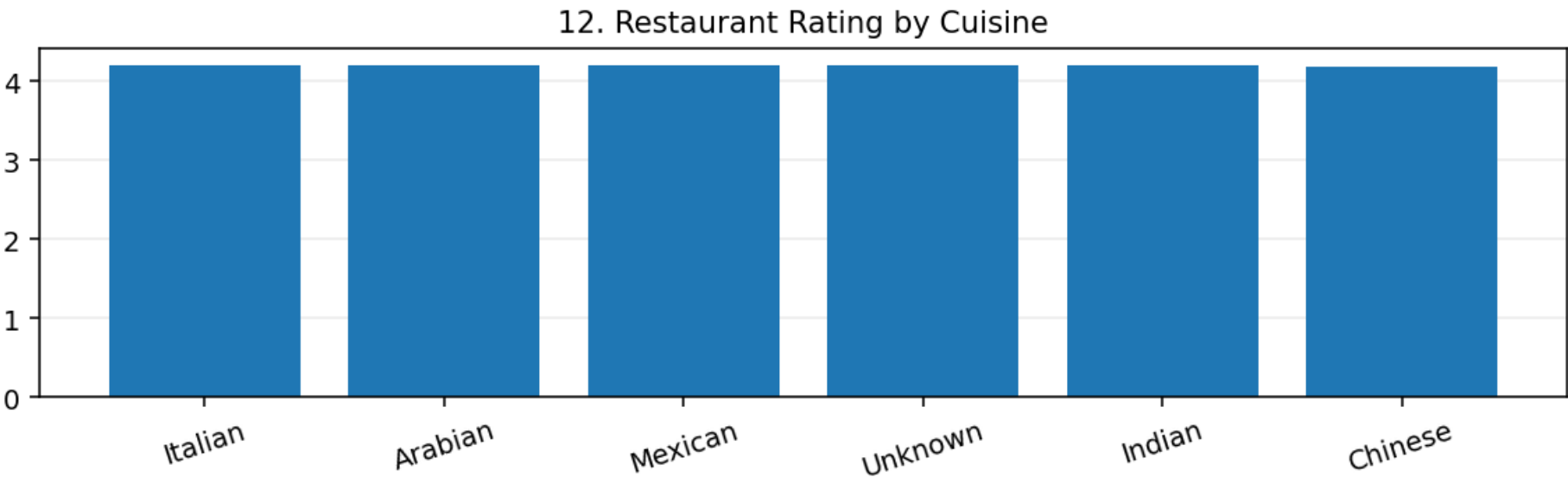
SQL Query

```
SELECT COALESCE(NULLIF(TRIM(Cuisine_Type),''),'Unknown / Missing') AS Cuisine, COUNT(Order_ID) AS Total_Orders, ROUND(AVG(Restaurant_Rating),2) AS Avg_Rating, MIN(Restaurant_Rating) AS Min_Rating, MAX(Restaurant_Rating) AS Max_Rating FROM food_orders GROUP BY Cuisine ORDER BY Avg_Rating DESC;
```

Data — actual supplied output

Cuisine	Orders	Avg Rating	Min	Max
Italian	16,519	4.21	3	5
Arabian	16,658	4.20	3	5
Mexican	16,602	4.20	3	5
Unknown / Missing	16,885	4.20	3	5
Indian	16,685	4.20	3	5
Chinese	16,651	4.19	3	5

Visual



Insights

- Italian has the highest average rating (4.21).
- Chinese has the lowest listed average (4.19).
- The 0.02-point spread is small; differences should not be overstated.

13. Discount vs Order Value

Objective: Compare average final order value across discount amounts.

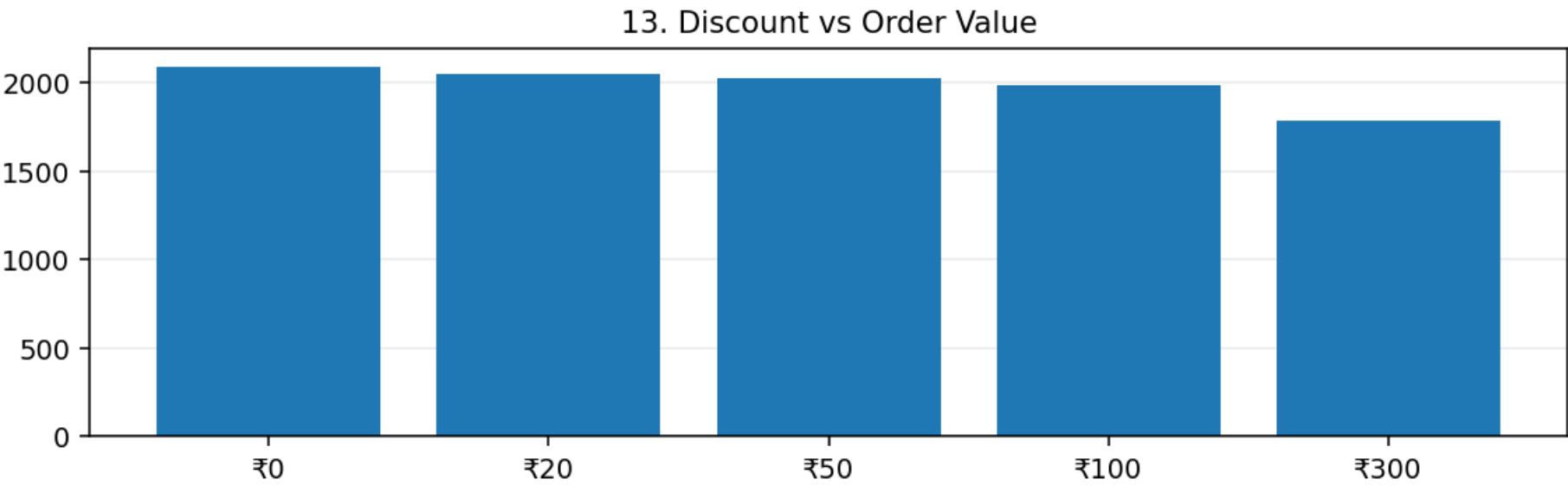
SQL Query

```
SELECT Discount_Applied, COUNT(Order_ID) AS Total_Orders, ROUND(AVG(Final_Amount),2) AS Avg_Order_Value, SUM(Final_Amount) AS Total_Revenue FROM food_orders GROUP BY Discount_Applied ORDER BY Discount_Applied;
```

Data — actual supplied output

Discount (■)	Orders	Avg Order (■)	Revenue (■)
0	33,365	2,088.29	46,715,054
20	16,569	2,048.23	22,596,117
50	16,690	2,027.78	22,459,738
100	16,776	1,983.29	22,101,758
300	16,600	1,784.91	19,725,053

Visual



Insights

- Average order value falls across the higher discount groups in this dataset.
- No-discount orders have the highest average order value: ■2,088.29.
- This association does not prove discounts cause lower order values; compare basket mix and customer behaviour.

14. Peak-Hour Flag Analysis

Objective: Compare revenue by peak-hour flag category.

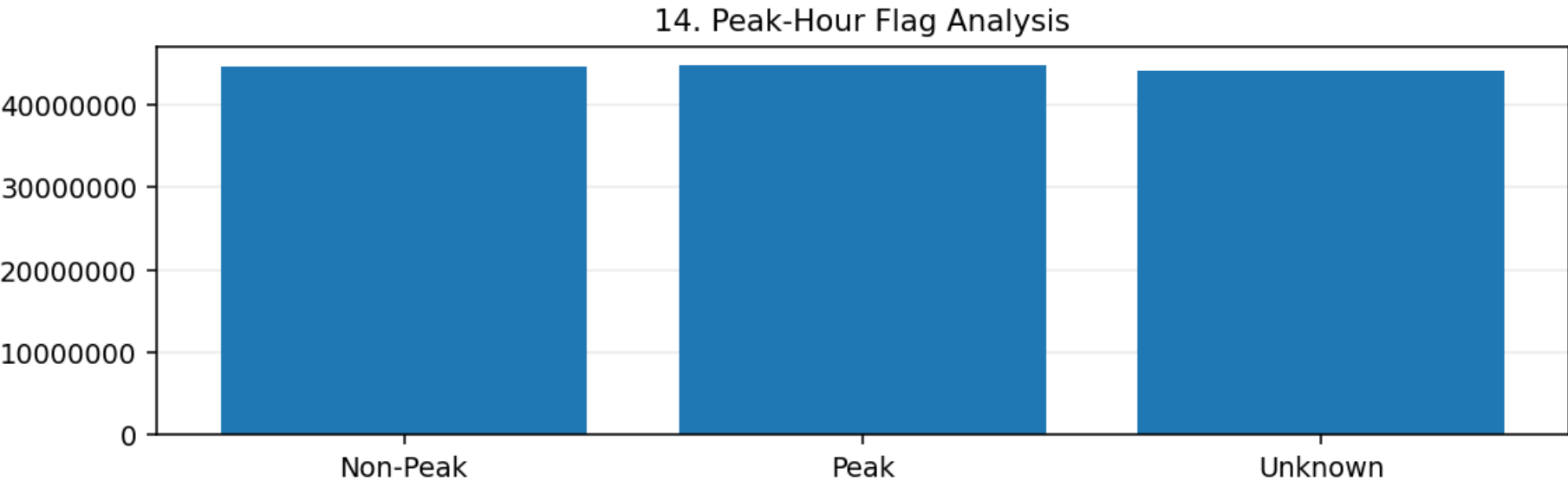
SQL Query

```
SELECT CASE WHEN Peak_Hour=1 THEN 'Peak Hour' WHEN Peak_Hour=0 THEN 'Non-Peak Hour' ELSE 'Unknown / Missing' END AS Peak_Category, COUNT(Order_ID) AS Total_Orders, SUM(Final_Amount) AS Total_Revenue, ROUND(AVG(Final_Amount),2) AS Avg_Order_Value FROM food_orders GROUP BY Peak_Category ORDER BY Total_Revenue DESC;
```

Data — actual supplied output

Category	Orders	Revenue (■)	Avg Order (■)
Non-Peak Hour	33,585	44,693,354	2,004.91
Peak Hour	33,453	44,774,727	2,003.25
Unknown / Missing	32,962	44,129,639	2,003.16

Visual



Insights

- Peak Hour revenue is ■44,774,727, slightly above Non-Peak.
- Average order values are very similar across groups.

- Peak_Hour is a binary flag, not the actual clock hour.

15. Top Restaurants by Revenue

Objective: Rank restaurants by delivered-order revenue.

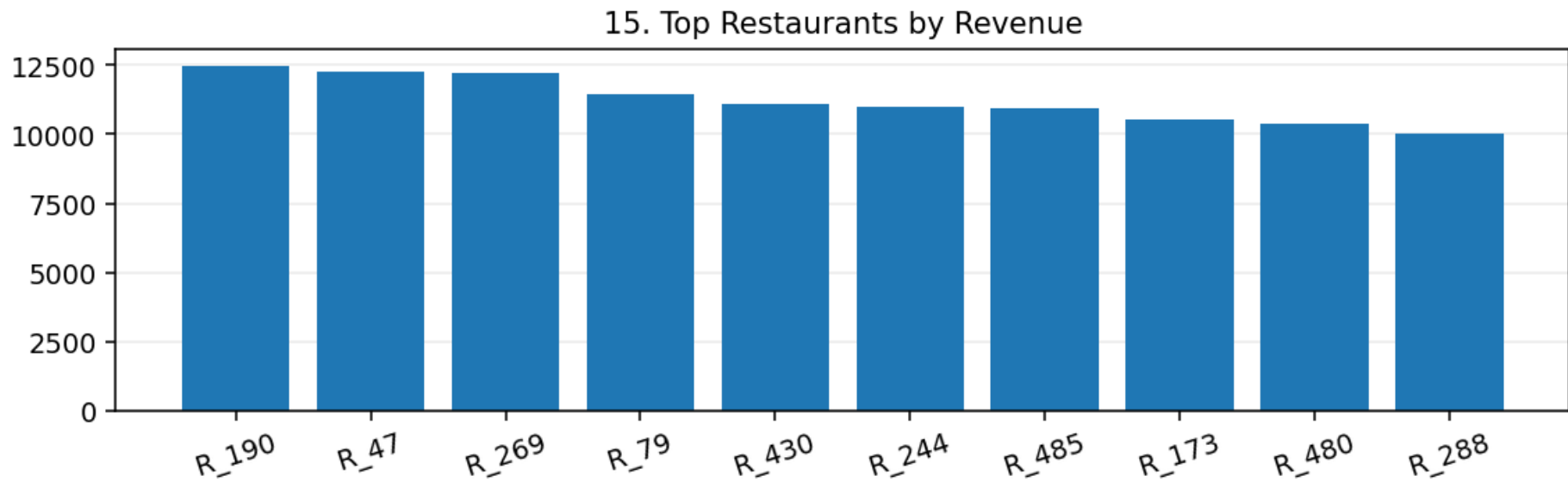
SQL Query

```
SELECT Restaurant_ID, Restaurant_Name, COUNT(Order_ID) AS Total_Orders, SUM(Final_Amount) AS Total_Revenue, ROUND(AVG(Final_Amount),2) AS Average_Order_Value FROM
food_orders WHERE Order_Status='Delivered' GROUP BY Restaurant_ID, Restaurant_Name ORDER BY Total_Revenue DESC LIMIT 10;
```

Data — actual supplied output

Restaurant ID	Restaurant Name	Orders	Revenue (■)	Avg Order (■)
RES997	Restaurant_190	3	12,464	4,154.67
RES630	Restaurant_47	4	12,288	4,096.00
RES180	Restaurant_269	3	12,229	4,076.33
RES546	Restaurant_79	3	11,435	3,811.67
RES385	Restaurant_430	3	11,116	3,705.33
RES169	Restaurant_244	3	10,972	3,657.33
RES220	Restaurant_485	3	10,947	3,649.00
RES177	Restaurant_173	3	10,553	3,517.67
RES340	Restaurant_480	3	10,388	3,462.67
RES815	Restaurant_288	3	10,035	3,345.00

Visual



Insights

- Restaurant_190 leads with 12,464 from 3 delivered orders.
- Restaurant_47 has the highest order count in the list (4).
- Each restaurant has only 3–4 orders; interpret the ranking cautiously.